

USC Viterbi School of Engineering

DSCI 510: Principles of Programming for Data Science
Units: 4

Fall 2026

32433 & 32405: OHE 122, Tuesday, 4-5:50pm, Lecture
32434 & 32428: OHE 122, Thursday, 4-5:50pm, Lab

Instructor: Wensheng Wu
Office Hours: 10-11am Tuesday and Thursday
(Details will be posted on the course web site)
Email: wenshenw@usc.edu

TAs: To be announced on course web site.
Office Hours: TBD

A. Catalogue Course Description

Programming in Python for retrieving, searching and analyzing data from the Web.
Learning to manipulate large data sets.

B. Expanded Course Description

The course is intended to provide students with a comprehensive introduction to Python programming language, preparing students for more advanced courses in the Data Science program such as DSCI 551 (data management), DSCI 552 (machine learning), DSCI 553 (data mining), and DSCI 554 (data visualization). Students might or might not have prior programming experiences, prior to taking DSCI 510.

The course develops Python programming skills from fundamental language constructs through data structures, object-oriented programming, numerical computing, data analysis, and machine learning. Emphasis is placed on computational thinking, program design, and understanding the principles underlying commonly used Python abstractions.

Students will also complete an individual or team project that applies course concepts to the design and implementation of a substantial Python-based application or system, culminating in a demonstration and final report.

C. Learning Objectives

By the end of the course, students will be able to:

- ***Write, test, and debug Python programs*** using core language constructs, functions, modules, exceptions, and standard data types.
- ***Select and apply appropriate data structures and algorithms*** while reasoning about their functionality and computational efficiency.
- ***Design object-oriented programs*** using classes, inheritance, composition, and Python's object model.
- ***Explain key mechanisms underlying Python programs***, including object creation, attribute lookup, method invocation, and method resolution.
- ***Use NumPy and pandas for numerical computing and data analysis***, including array operations, indexing, filtering, grouping, aggregation, and data transformation.

- ***Use PyTorch to construct and train basic machine-learning models***, including automatic differentiation, neural-network layers, loss functions, and gradient-based optimization.
- ***Design, implement, and communicate a substantial Python project***, individually or as part of a team, integrating concepts and techniques from the course.

D. Course Structure and Notes

The course will be conducted through lectures and labs, with labs providing guided exercises and implementation practice that reinforce the lecture material. Active student participation is required. All course materials, including lecture notes, assignments, project materials, and other resources, will be posted on the course website.

E. Technological Proficiency and Hardware/Software Required

Students should have access to a laptop or desktop computer on which they can install and run the software required to complete course assignments and the project. Students should be comfortable installing software, managing files, and using a web browser and development environment for Python programming.

F. Recommended Readings and Supplementary Materials

The course will use the following texts as references. All texts are available online at no additional cost to USC students through the [O'Reilly Learning Platform](#). To access the materials, select **University of Southern California** as your institution and sign in using your USC email address.

- [PCC] Eric Matthes. [Python Crash Course](#), 3rd Edition, 2023.
- [Fluent] Luciano Ramalho. [Fluent Python](#), 2nd Edition, 2022.
- [Handbook] Jake VanderPlas, [Python Data Science Handbook, 2nd Edition](#), 2022.

G. Grading Scheme

Homework assignments: There will be 10 homework assignments, as indicated in the course schedule below. All homework assignments must be completed **individually**. Each assignment will typically be graded on a scale of 0–100, and specific requirements and grading guidelines will be provided with each assignment.

Exams: There will be two exams, as indicated in the course schedule below. All exams will be closed-book and closed-notes. On-campus students will take the exams in the regular classroom, while DEN students will take them remotely at the same scheduled time. DEN students are required to join the class Zoom meeting with their cameras on for virtual proctoring.

Course project: Students are expected to complete a term project that applies Python programming concepts and techniques covered in the course. The project may be completed individually or as part of a team. Project requirements, guidelines, and deliverables will be announced separately.

Grade breakdown:

Component	Weight
Homework	30%
Exam 1	25%
Exam 2	25%
Course Project	20%
Total	100%

Letter grades: Letter grades will range from A through F and will be assigned according to the following cutoffs:

Grade Range	Letter Grade	Grade Range	Letter Grade
93–100	A	73–<77	C
90–<93	A–	70–<73	C–
87–<90	B+	67–<70	D+
83–<87	B	63–<67	D
80–<83	B–	60–<63	D–
77–<80	C+	Below 60	F

This course uses **absolute grading; no curve will be applied**. Final grades will be determined strictly according to the stated grading criteria and are not negotiable. Every point counts, and grades will not be rounded up. Requests for grade rounding, exceptions to the grading policy, or other special treatment will not be considered. Students are expected to review their graded work promptly and raise any legitimate grading errors through the established regrade process.

H. Course Policies

- **Coursework Deadlines and Late Submissions**

Coursework, including homework assignments and project deliverables, is due by **11:59 p.m. on the specified due date** and must be submitted through the course website as announced. **Late submissions will not be accepted.**

Students are responsible for ensuring that they have a stable network connection, that their submissions are completed before the deadline, and that the correct files have been submitted. **Technical difficulties, network connectivity problems, and submission of incorrect files will not be accepted as grounds for late submission or deadline extensions.** Students are strongly encouraged to submit coursework well before the deadline and verify their submissions after uploading.

- **Exams and Deadline Extensions**

Makeup exams and extensions of coursework deadlines will be considered only in cases of **documented medical emergencies or other circumstances for which**

accommodations are required by university policy. Appropriate documentation may be required.

Requests must be made **before the applicable exam or coursework deadline.** For foreseeable circumstances, requests for makeup exams must be submitted at least **one week in advance**, and requests for coursework extensions must be submitted at least **48 hours in advance.**

Conflicts arising from interviews, job fairs, travel, or other personal commitments do not qualify for makeup exams or deadline extensions. Students are responsible for planning their schedules to avoid conflicts with class meetings, exams, and coursework deadlines. **No exceptions will be made except as required by university policy.**

Students who require accommodations should contact the OSAS office as early as possible and provide their accommodation letters to the teaching staff before the relevant exams or coursework deadlines. Accommodations cannot be applied retroactively. Students using the OSAS Testing Center must schedule their exams at least ****one week before**** the exam date.

- **Regrading Requests**

Requests for regrading must be submitted to the TAs, or through the procedure announced by the instructional staff, within **one week after the relevant grades or solutions have been posted**, unless otherwise announced. Regrading requests must identify a specific grading issue or error. Grades become final after the regrading period has ended.

A regrade request may result in the re-evaluation of the entire assignment or exam, not only the specific question in dispute. Consequently, a regrade may result in a lower score if previously overlooked grading errors are discovered.

Unless an answer is entirely correct and contains no syntax or semantic errors, the awarding of partial credit—and the amount awarded, if any—is at the grader's discretion.

I. Course Schedule: A Weekly Breakdown (may be revised when the course progresses)

Week	Topic	Recommended Readings	HW & Project
1 (8/24)	Python foundations: • Variables and data types; expressions, statements, and operators; conditionals and loops; input/output and file I/O	• PCC: Ch. 1–2, 5, 7, 10	
2 (8/31)	Functions and Program Structure: • Functions; parameters and arguments; scope; recursion; modules and imports	• PCC: Ch. 8 • Fluent: Ch. 7, 9 (selected)	HW1
3 (9/7)	Collections, Iteration, and Exceptions:	• PCC: Ch. 3–4, 6,	HW2

	Lists, tuples, dictionaries, and sets; comprehensions; iteration; iterators/generators; exception handling	10 Fluent: Ch. 2–3, 17 (selected)	
4 (9/14)	Sequences and Algorithmic Efficiency: Sequence operations; dynamic arrays; indexing and slicing; sorting; algorithmic complexity	Fluent: Ch. 2 (selected)	HW3
5 (9/21)	Review Exam 1 on 9/24, Thursday		
6 (9/28)	Dictionaries, Sets, Searching, and Priority Queues: Hashing and hash tables; dictionaries and sets; binary search; heaps and priority queues; complexity	Fluent: Ch. 3 (selected)	HW4
7 (10/5)	<i>No class on 10/8, Fall Recess.</i> Object-Oriented Programming: Classes and objects; attributes and methods; object creation; special methods; operator overloading and callable objects	PCC: Ch. 9 Fluent: Ch. 1, 7, 11 (selected)	HW5 Project proposal due
8 (10/12)	Inheritance and Python's Object Model: Inheritance and overriding; multiple inheritance; MRO and super(); attribute lookup; bound methods	Fluent: Ch. 14, 23 (selected)	HW6
9 (10/19)	Advanced Object-Oriented Programming: Classes as objects; object and type; descriptors; properties; class and static methods	Fluent: Ch. 22–24	HW7
10 (10/26)	Review Exam 2 on 10/29, Thursday		
11 (11/2)	Numerical Computing with NumPy: Multidimensional arrays; indexing and slicing; views and reshape; vectorized operations; broadcasting and reductions	Handbook: Part 2 (selected topics)	HW8
12 (11/9)	Data Analysis with pandas: Series and DataFrame; labeled and positional indexing; selection and boolean filtering; sorting and querying; grouping and aggregation	Handbook: Part 3 (selected topics)	HW9
13 (11/16)	Machine Learning with PyTorch: Tensors; computational graphs and autograd; neural networks; loss functions; gradient-based training	Selected PyTorch documentation /tutorials	HW10
14 (11/23)	<i>No class on 11/26, Thanksgiving.</i> Project demo		
15 (11/30)	Project demo		

Final week	Project final report due on 12/10, Thursday		
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Statement on Academic Conduct and Support Systems

Academic Integrity:

The University of Southern California is a learning community committed to developing successful scholars and researchers dedicated to the pursuit of knowledge and the dissemination of ideas. Academic misconduct, which includes any act of dishonesty in the production or submission of academic work, comprises the integrity of the person who commits the act and can impugn the perceived integrity of the entire university community. It stands in opposition to the university's mission to research, educate, and contribute productively to our community and the world.

All students are expected to submit assignments that represent their own original work, and that have been prepared specifically for the course or section for which they have been submitted. You may not submit work written by others or "recycle" work prepared for other courses without obtaining written permission from the instructor(s).

Other violations of academic integrity include, but are not limited to, cheating, plagiarism, fabrication (e.g., falsifying data), collusion, knowingly assisting others in acts of academic dishonesty, and any act that gains or is intended to gain an unfair academic advantage.

The impact of academic dishonesty is far-reaching and is considered a serious offense against the university. All incidences of academic misconduct will be reported to the Office of Academic Integrity and could result in outcomes such as failure on the assignment, failure in the course, suspension, or even expulsion from the university.

For more information about academic integrity see [the student handbook](#) or the [Office of Academic Integrity's website](#), and university policies on [Research and Scholarship Misconduct](#).

Please ask your instructor if you are unsure what constitutes unauthorized assistance on an exam or assignment, or what information requires citation and/or attribution.

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University's educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

[Counseling and Mental Health](#) - (213) 740-9355 – 24/7 on call

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

[988 Suicide and Crisis Lifeline](#) - 988 for both calls and text messages – 24/7 on call

The 988 Suicide and Crisis Lifeline (formerly known as the National Suicide Prevention Lifeline) provides free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week, across the United States. The Lifeline is comprised of a national network of over 200 local crisis centers, combining custom local care and resources with national standards and best practices. The new, shorter phone number makes it easier for people to remember and access mental health crisis services (though the previous 1 (800) 273-8255 number will continue to function indefinitely) and represents a continued commitment to those in crisis.

[Relationship and Sexual Violence Prevention Services \(RSVP\)](#) - (213) 740-9355(WELL) – 24/7 on call

Free and confidential therapy services, workshops, and training for situations related to gender- and power-based harm (including sexual assault, intimate partner violence, and stalking).

[Office for Equity, Equal Opportunity, and Title IX \(EEO-TIX\)](#) - (213) 740-5086

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

[Reporting Incidents of Bias or Harassment](#) - (213) 740-5086 or (213) 821-8298

Avenue to report incidents of bias, hate crimes, and microaggressions to the Office for Equity, Equal Opportunity, and Title for appropriate investigation, supportive measures, and response.

[The Office of Student Accessibility Services \(OSAS\)](#) - (213) 740-0776

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

[USC Campus Support and Intervention](#) - (213) 740-0411

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

[Diversity, Equity and Inclusion](#) - (213) 740-2101

Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

[USC Emergency](#) - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

[USC Department of Public Safety](#) - UPC: (213) 740-6000, HSC: (323) 442-1200 – 24/7 on call

Non-emergency assistance or information.

[Office of the Ombuds](#) - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

[Occupational Therapy Faculty Practice](#) - (323) 442-2850 or otfp@med.usc.edu

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.